Source and gain ambiguity

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3. gain-shift ambiguity with arbitrary common source covariance

Let L be a **connected real symmetric weighted graph Laplacian** on n *already labeled* sites, with P0=J/n, Q=I-P0, W_s=sLe^{-sL}. Take distinct **known** scales 0<s<t, a **single shared** zero-mean random source x with covariance Σ strictly positive on the mean-zero subspace, and positive scalar detector gains g_s,g_t:

y_s=g_s W_s x, y_t=g_t W_t x.

The jointly measured covariances are

C_st=g_s g_t W_s Σ W_t, C_tt=g_t² W_t Σ W_t.

All operators W_s and W_t vanish on P0 and are invertible on Q. On Q, C_st C_tt^+ = (g_s/g_t) W_s W_t^{-1}, independent of the generally **noncommuting** covariance Σ. Thus

C_st C_tt^+ = (g_s/g_t) (s/t) exp[(t-s)L]Q.

Consequently the nominal logarithmic reconstruction without calibrated relative gain returns

\hat L = (1/(t-s)) log[P0+(t/s) C_st C_tt^+] = L+αQ,

where α=log(g_s/g_t)/(t-s).

This is an **exact one-parameter ambiguity**, not an additive noise approximation. Unknown Σ by itself cancels for paired measurements; unknown relative gain does not. The identity fails if cross-scale observations do not share the same x, if Σ loses rank on Q, or if the readout covariances contain uncorrected independent additive detector noise.

A conditional self-calibration that weakens the gain premise

A weighted graph Laplacian has offdiagonal entries L_ij=-w_ij ≤ 0. If the candidate is independently known to have **at least one absent edge** in the measured site basis, then max_{i≠j} L_ij=0. As Q_ij=-1/n for i≠j,

\hat L_ij=L_ij-α/n.

Therefore

α=-n max_{i≠j}\hat L_ij,

L=\hat L-αQ, and g_s/g_t=exp[(t-s)α].

The location of the missing edge need not be known. If an approximate shifted generator has entrywise error at most δ, the extracted α has error at most nδ, and the final recovered generator has entrywise error at most nδ. This controls errors **after** the logarithmic inversion; it is not a uniform stability bound from noisy covariance measurements, whose inversion may be severely ill-conditioned. The existence of the missing edge must, however, be justified independently; otherwise the inference has simply inserted sparsity into the premises. This statement is false as unconditional graph reconstruction: for a fully connected positive-weight graph, adding a small complete-graph term generates alternative valid graph Laplacians.

Adversarial source-and-gain transformation

For any c>0, L_c=L+cQ is a valid connected graph Laplacian adding weight c/n to *every* distinct pair. With

T_c=P0 + L(L+cQ)^{-1}_Q Q, x'=T_c x, g'_s=g_s e^{sc},

one has exactly g'_s W_s(L_c)x'=g_s W_s(L)x for **all scales** and individual realizations. Hence if the source and gain laws are insufficiently calibrated, sparse and complete graphs can be observationally identical. The conditional sparsity recovery above **rejects** the complete augmentation only because its prior forbids the augmented model.

Numerical checks

Constructed both 16-vertex rook/Shrikhande Laplacians and a nonwhite, noncommuting positive-definite Σ. Used scales s=log2/4, t=.31, gains g_s=1.73, g_t=.67. The recovered α is 6.93860535328…; the exact target is log(1.73/.67)/(.31-log2/4). Reconstructed each L by the independent no-missing-edge-coordinate self-calibration with entrywise errors 1.25e-13 (rook), 2.26e-13 (Shrikhande). With complete-graph augmentation c=2.3 and transformed sources/gains, four separate filter scales agree to <2.5e-15 in maximum matrix entry. These are floating-point corroborations of the exact formulas, not formal error bars or physical data. The test script and JSON output are retained. A separate verification program constructs both graphs, enumerates all induced three-port types and four-cliques, compares noncommuting three-port control sequences, and verifies an exact π-flux four-cycle caging identity. All assertions pass locally, with three-port controlled response difference 7.33e-16, Krylov ranks 7 in both chosen triangle-port experiments, and π-flux opposite-corner transfer below 6.1e-17.

Prior art boundary: blind graph deconvolution, spectrum/graph learning from structured signals, and system-identification similarity classes are established fields (e.g. Hammond et al. 2011; Ramírez et al. 2017; Burgarth and Yuasa 2014). These conditional formulas are a working synthesis and do not establish publication novelty.

4. Operational readout of full commutation: a simple robustness calculation

Let A and B be Pauli observables with A²=B²=I. After a projective measurement of B, apply U(θ)=exp(iθA), then measure B again. Let b,b’ be the consecutive ideal outcomes. For commuting A,B, E[bb']=1. For anticommuting A,B, E[bb']=cos(2θ). At θ=π/2 they are +1 and -1, respectively, irrespective of the state. This distinguishes the two models without interpreting a global -I phase as an observable by itself.

Under the **specific** error model in which the two recorded bits are independently flipped with probability e, and θ=π/2+δ, the observed contrast is

Δ_corr=2(1-2e)² cos² δ.

Thus a simple instrument noise model retains a finite discrimination signal whenever e<1/2 and |δ|<π/2. This is not a guarantee for arbitrary invasive/nonprojective measurements, source drift, or pulse crosstalk. For e=.08, δ=.07, the predicted contrast is 1.40429640693, checked by full two-qubit matrices. The required B in the dressed model is itself a nonlocal Pauli string under the original site interpretation, so access to its projective measurement is a substantial operational input.

If a faithful state ρ on dimension d has known smallest eigenvalue r>0, then Tr[ρ C†C] ≥ r ||C||_op² for C=[A,B]. A measured small commutator *mean-square* controls an operator-norm cross-commutator only with this faithfulness bound. The weaker expected commutator Tr(ρ[A,B])≈0 does not provide it. If r=0 or scales to zero with d, no uniform operator-level guarantee follows.